

# AMC 10 Crash Course • Block 1 • Algebra • Lessons 1.1–1.5

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

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## Lesson 1.1. Percents, Ratios, Averages, Motion

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### THEORY

**Percents as multipliers.** A 20% increase means multiplying by 1.2; a 25% decrease means multiplying by 0.75. Successive changes multiply: +20% followed by −25% gives  $1.2 \cdot 0.75 = 0.9$ , that is −10%. Never add percents taken of different bases.

A change of **p%** = multiplication by **(1 + p/100)**. A chain of changes = the product of the multipliers.

**Ratios via parts.** If two quantities are in the ratio 3 : 4, let  $k$  be the size of one part: the quantities are  $3k$  and  $4k$ . Chained ratios are linked through the common quantity: from  $a : b = 2 : 3$  and  $b : c = 4 : 5$ , scale  $b$  to 12 to get  $a : b : c = 8 : 12 : 15$ .

**Averages.** Mean = sum / count; nearly every average problem is solved by working with the **sum**: find the sum before and after the change. A weighted average lies between the group means, closer to the larger group.

**Motion.** Distance = speed  $\times$  time. Average speed is the TOTAL distance over the TOTAL time, not the average of speeds: over equal distances it equals  $\frac{2v_1v_2}{v_1 + v_2}$  and is always less than the arithmetic mean.

### WORKED EXAMPLES

#### EXAMPLE 1 • PERCENTS

A price first rose by 25%, then fell by 20%. How does the final price compare to the original?

→ Multipliers:  $1.25 \cdot 0.8 = 1$ . The price is unchanged. +25 and −20 look like they should not cancel — but as multipliers they do:  $5/4 \cdot 4/5 = 1$ . Fractions are faster than decimals here.

#### EXAMPLE 2 • RATIOS

In a class the ratio of boys to girls is 3 : 4. After 6 more boys join, the ratio becomes 9 : 8. How many students are in the class now?

→ Originally  $3k$  and  $4k$ . Now:  $\frac{3k+6}{4k} = \frac{9}{8}$ . Cross-multiply:  $24k + 48 = 36k$ , so  $k = 4$ . The class now has  $(12 + 6) + 16 = \mathbf{34}$  students. Check:  $18 : 16 = 9 : 8$ . Correct.

#### EXAMPLE 3 • AVERAGE SPEED

You drive to a town at 60 mph and return along the same road at 40 mph. What is the average speed for the whole trip?

→ Let the one-way distance be  $d$ . Time:  $d/60 + d/40 = d \cdot 5/120 = d/24$ . Average =  $2d \div (d/24) = \mathbf{48 \text{ mph}}$ . Not 50! The formula  $2 \cdot 60 \cdot 40 / (60 + 40)$  gives the same instantly. On the AMC, the arithmetic mean of the speeds almost always appears among the choices as a trap.

#### EXAMPLE 4 • WEIGHTED AVERAGE

The girls' average score is 90, the boys' is 80, and the class average is 84. What fraction of the class are boys?

→ Let the boys' fraction be  $m$ . Then  $80m + 90(1 - m) = 84$ , so  $90 - 10m = 84$  and  $m = 0.6$ : **60%**. Quick lever check: 84 is 4 away from 80 and 6 away from 90; weights are inverse to distances,  $6 : 4 = 3 : 2$  in favor of the boys.

### INDEPENDENT SET

1. 20% of 30% of a number equals 12. Find the number.
2. An item's price dropped by 50%. By what percent must it rise to return to the original price?

3. Two liters of a 10% acid solution are mixed with three liters of a 20% solution. What is the concentration of the mixture?
4.  $a : b = 2 : 3$  and  $b : c = 4 : 5$ . Find  $a : c$ .
5. The average of seven numbers is 15. An eighth number is added, and the average becomes 16. What number was added?
6. A car drives the first half of a route at 30 mph and the second half at 60 mph. What is its average speed?
7. A train 880 feet long passes a pole in 10 seconds. What is its speed in miles per hour?
8. A salary was raised by 10%, and a year later by another 10%. By what percent did it grow over the two years?
9. ★ A container holds a 40% acid solution. A quarter of the contents is removed and replaced with water; then a quarter is removed and replaced with water again. What is the concentration now?

## Lesson 1.2. Equations and Systems, Work and Mixtures, Sequences

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### THEORY

**Linear equations and systems.** Expand, collect  $x$  on the left and numbers on the right, divide. In a two-equation system, solve for the variable whose coefficient is 1 and substitute into the other equation. Plug every answer back in: the check takes ten seconds and catches half of all mistakes.

**Work problems.** The key is the rate: the fraction of the job done per hour. A pipe that fills a pool in 6 hours does  $1/6$  of the job per hour. Rates add: a “6-hour” pipe and a “3-hour” pipe together do  $1/6 + 1/3 = 1/2$  of the job per hour, i.e. 2 hours for the whole job.

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| $\text{Work per hour} = \frac{1}{\text{time alone}}. \text{ Joint time} = \frac{1}{\text{sum of rates}}.$ |
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**Mixtures.** Track the pure substance, not the percents: how many liters of pure juice/acid there were and there are. Added water increases the volume but not the substance.

**Sequences.** Arithmetic:  $a_n = a_1 + (n - 1)d$ ;  $\text{sum} = \frac{a_1 + a_n}{2} \cdot n$  (the mean of the first and last, times the count).

Geometric:  $a_n = a_1 r^{n-1}$ ;  $\text{sum} = a_1 \frac{r^n - 1}{r - 1}$  (for  $r \neq 1$ ).

### WORKED EXAMPLES

#### EXAMPLE 1 • SYSTEM

Three pens and two notebooks cost \$24; one pen and four notebooks cost \$18. How much does a pen cost?

→ From the second equation,  $p = 18 - 4n$ . Substitute:  $3(18 - 4n) + 2n = 24$ , so  $54 - 10n = 24$ ,  $n = 3$  and  $p = \$6$ . Check:  $3 \cdot 6 + 2 \cdot 3 = 24$ . Correct.

#### EXAMPLE 2 • WORK

One pipe fills a pool in 10 hours, another in 15. How long do they take together?

→ Rates:  $1/10 + 1/15 = 1/6$  of the pool per hour. Together: **6 hours**. The common trap “ $(10+15)/2$ ” is wrong: together is always faster than the faster pipe alone.

#### EXAMPLE 3 • MIXTURE

Five liters of juice have concentration 12%. How much water must be added to get 8%?

→ Pure juice:  $0.12 \cdot 5 = 0.6$  L, and it does not change. New volume:  $0.6 / 0.08 = 7.5$  L. Add **2.5 liters**. The whole problem is one division if you track the substance, not the percents.

#### EXAMPLE 4 • SEQUENCES

Find the sum  $3 + 6 + 12 + \dots + 384$ .

→ A geometric sequence:  $a_1 = 3$ ,  $r = 2$ . The last term is  $384 = 3 \cdot 2^7$ , so there are 8 terms. Sum  $= 3(2^8 - 1)/(2 - 1) = 3 \cdot 255 = 765$ . Quick check for  $r = 2$ : the sum is twice the last term minus the first:  $768 - 3 = 765$ .

#### INDEPENDENT SET

1. Solve the equation  $5(x - 2) = 3x + 8$ .
2. The sum of two numbers is 40, and one is 6 more than the other. Find their product.
3. A painter paints a fence in 4 hours, an apprentice in 12. How long will they take together?
4. Two liters of a 30% solution are mixed with four liters of a 15% solution. What is the concentration of the mixture?
5. In an arithmetic sequence,  $a_1 = 7$  and  $d = 4$ . Find  $a_{25}$ .
6. Find the sum of the first 30 odd positive integers:  $1 + 3 + 5 + \dots + 59$ .
7. In a geometric sequence,  $a_3 = 18$  and  $a_5 = 162$ . Find  $a_1$ .
8. Maya solves 2 problems per minute; Ben solves 3. Ben starts 5 minutes after Maya. How many minutes after Maya starts will they have solved the same number of problems?
9. ★ The sum of the first  $n$  terms of a sequence is  $S_n = 3n^2 + 2n$ . Find the tenth term.

### Lesson 1.3. Quadratics: Roots, Vieta's Formulas, the Vertex

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#### THEORY

**Roots.** Try factoring first:  $x^2 - 7x + 12 = (x - 3)(x - 4)$ . If it does not factor in ten seconds, use the discriminant:  $D = b^2 - 4ac$ , roots  $(\frac{-b \pm \sqrt{D}}{2a})$ .  $D > 0$  — two roots,  $D = 0$  — one (a double root),  $D < 0$  — no real roots.

**Vieta's formulas** for  $x^2 + px + q$ : sum of roots  $= -p$ , product  $= q$ . The competition payoff: you can learn everything about the roots without ever finding them.

**Symmetric expressions via Vieta.**  $r^2 + s^2 = (r + s)^2 - 2rs$ ;  $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$ ;  $1/r + 1/s = (r + s)/rs$ . These three cover most problems.

**Vertex.** The parabola  $ax^2 + bx + c$  has its vertex at  $x = -b/2a$  — the minimum point ( $a > 0$ ) or maximum point ( $a < 0$ ). “Find the least value” problems are almost always a vertex substitution. And the sign: for  $a > 0$ , the quadratic is negative strictly between its roots and positive outside them — that is all you need for quadratic inequalities.

#### WORKED EXAMPLES

##### EXAMPLE 1 • FACTORING

Solve  $x^2 - 7x + 12 = 0$ .

→ Look for two numbers with sum 7 and product 12: these are 3 and 4. Roots: **3 and 4**. No discriminant needed: factoring is faster and avoids arithmetic slips.

### EXAMPLE 2 • VIETA

Let  $r$  and  $s$  be the roots of  $x^2 - 5x + 3 = 0$ . What is  $r^2 + s^2$ ?

→ By Vieta's formulas,  $r + s = 5$ ,  $rs = 3$ . Then  $r^2 + s^2 = 25 - 2 \cdot 3 = \mathbf{19}$ . The roots themselves are irrational, and finding them wastes two minutes.

### EXAMPLE 3 • VERTEX

Find the least value of  $x^2 - 6x + 11$ .

→ Vertex at  $x = 6/2 = 3$ . Value:  $9 - 18 + 11 = \mathbf{2}$ . Alternatively, complete the square:  $(x - 3)^2 + 2 \geq 2$ , which shows the minimum is 2, at  $x = 3$ .

### EXAMPLE 4 • PARAMETER

For which  $k$  does  $x^2 + kx + 9 = 0$  have exactly one root?

→ Exactly one root  $\Leftrightarrow D = 0$ :  $k^2 - 36 = 0$ , so  $k = \mathbf{6 \text{ or } -6}$ . The classic way to lose a point here is forgetting the negative value.

## INDEPENDENT SET

1. Solve  $x^2 = 5x$ . Give all roots.
2. Find the largest root of  $x^2 - 2x - 15 = 0$ .
3. Roots  $r$  and  $s$  satisfy  $r + s = 6$  and  $rs = 4$ . What is  $1/r + 1/s$ ?
4. Let  $r$  and  $s$  be the roots of  $x^2 - 4x + 1 = 0$ . What is  $r^3 + s^3$ ?
5. Find the greatest value of  $-x^2 + 8x - 7$ .
6. The number 2 is a root of  $x^2 - 6x + c = 0$ . Find the other root.
7. How many integers satisfy  $x^2 - 5x + 6 < 0$ ?
8. A quadratic with integer coefficients has roots  $3 + \sqrt{2}$  and  $3 - \sqrt{2}$ . What is the product of the roots?
9. ★ Let  $r$  and  $s$  be the roots of  $x^2 - 3x - 2 = 0$ . A quadratic of the form  $x^2 + px + q = 0$  has roots  $r + 1$  and  $s + 1$ . What is  $q$ ?

## Lesson 1.4. Identities and Tricks: Squares, Cubes, Telescoping

### THEORY

**Three identities to know on sight.**  $(a \pm b)^2 = a^2 \pm 2ab + b^2$ ; difference of squares  $a^2 - b^2 = (a - b)(a + b)$ ; sum and difference of cubes  $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ . The difference of squares is the biggest computational shortcut:  $101^2 - 99^2 = 200 \cdot 2$ .

If  $s = x + 1/x$ , then  $x^2 + 1/x^2 = s^2 - 2$ , and  $x^3 + 1/x^3 = s^3 - 3s$ . Raise what you know to a power and subtract the extra.

**Telescoping.** A long sum or product where neighboring terms cancel:  $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$  — everything inside collapses, leaving the first and last pieces. If you see a sum of a hundred fractions, look for telescoping.

**Grouping.**  $ab + a + b + 1 = (a + 1)(b + 1)$ . Adding one and factoring is a standard move in integer problems.

## WORKED EXAMPLES

### EXAMPLE 1 • DIFFERENCE OF SQUARES

Compute  $101^2 - 99^2$  without long multiplication.

→  $(101 - 99)(101 + 99) = 2 \cdot 200 = \mathbf{400}$ . On the AMC such computations are embedded in bigger problems: recognize the difference of squares instantly.

### EXAMPLE 2 • $X + 1/X$

Given  $x + 1/x = 3$ , find  $x^2 + 1/x^2$  and  $x^4 + 1/x^4$ .

→ Square it:  $9 = x^2 + 2 + 1/x^2$ , so  $x^2 + 1/x^2 = 7$ . Same move again:  $49 = x^4 + 2 + 1/x^4$ , so  $\mathbf{47}$ . Each squaring brings an extra 2 — subtract it.

### EXAMPLE 3 • TELESCOPING

Compute the sum  $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{99 \cdot 100}$ .

→ Each term:  $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ . The sum collapses:  $1 - 1/100 = \mathbf{99/100}$ .

### EXAMPLE 4 • GROUPING

Positive integers  $a$  and  $b$  satisfy  $ab + a + b = 63$ . What is the least possible value of  $a + b$ ?

→ Add 1:  $(a + 1)(b + 1) = 64$ . Ways to write 64 as a product of two factors  $\geq 2$ :  $2 \cdot 32$ ,  $4 \cdot 16$ ,  $8 \cdot 8$ . The sums  $a + b$ : 32, 18, 14. The least is  $\mathbf{14}$  (at  $a = b = 7$ ): with a fixed product, the sum is smallest when the factors are closest together.

## INDEPENDENT SET

1. Compute  $51^2 - 49^2$ .
2. Compute  $(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})$ .
3. Given  $x + 1/x = 6$ , find  $x^3 + 1/x^3$ .
4. Compute  $999 \cdot 1001$  without long multiplication.
5. Compute the product  $(1 - \frac{1}{2})(1 - \frac{1}{3}) \dots (1 - \frac{1}{100})$ .
6. Numbers  $a$  and  $b$  satisfy  $a + b = 7$  and  $ab = 10$ . Find  $a^3 + b^3$ .
7. Compute the sum  $\frac{1}{\sqrt{2} + 1} + \frac{1}{\sqrt{3} + \sqrt{2}} + \dots + \frac{1}{\sqrt{25} + \sqrt{24}}$ .
8. Compute  $\frac{2^{10} - 1}{2^5 - 1}$ .
9. ★ Compute  $(2 + 1)(2^2 + 1)(2^4 + 1)(2^8 + 1) + 1$ .

## Lesson 1.5. Absolute Value: Equations, Inequalities, Distances

### THEORY

**Absolute value is distance.**  $|x|$  is the distance from  $x$  to zero;  $|x - a|$  is the distance from  $x$  to the point  $a$ . Half of all absolute-value problems are solved faster by a number-line picture than by case analysis.

$$|f| = c \text{ (for } c > 0) \Leftrightarrow f = c \text{ or } f = -c. \text{ For } a > 0: |x| < a \Leftrightarrow -a < x < a; |x| > a \Leftrightarrow x < -a \text{ or } x > a.$$

**Sum of distances.**  $|x - a| + |x - b|$  is the total distance from  $x$  to two points. Between the points it is constant and equals  $|a - b|$ ; outside, it grows. For an odd number of points, the minimum of the sum of distances is at the middle (median) point.

**Nested absolute values.**  $||x| - 4| = 2$  unwraps layer by layer, outside in: first  $|x| - 4 = \pm 2$ , then each case separately. Care beats speed: count the roots at the end.

## WORKED EXAMPLES

### EXAMPLE 1 • EQUATION

Solve  $|2x - 3| = 7$ .

→  $2x - 3 = 7$  or  $2x - 3 = -7$ : roots **5 and -2**. Check by substitution:  $|7| = 7$ ,  $|-7| = 7$ . Correct.

### EXAMPLE 2 • SUM OF DISTANCES

What is the least value of  $|x - 1| + |x - 5|$ , and for which  $x$  is it attained?

→ This is the total distance from  $x$  to the points 1 and 5. Between them it is constant and equals the distance between the points: **4**, for every  $x$  in  $[1, 5]$ . No case analysis: one picture.

### EXAMPLE 3 • INEQUALITY

How many integers satisfy  $|x - 2| < 3$ ?

→  $-3 < x - 2 < 3$ , that is  $-1 < x < 5$ . Integers: 0, 1, 2, 3, 4 — **five**. A frequent mistake is counting the endpoints: with a strict inequality,  $-1$  and  $5$  are not included.

### EXAMPLE 4 • NESTED ABSOLUTE VALUES

How many solutions does  $||x| - 4| = 2$  have?

→ First layer:  $|x| - 4 = 2$  or  $-2$ , so  $|x| = 6$  or  $|x| = 2$ . Second layer:  $x = \pm 6$  and  $x = \pm 2$ . In total, **4 solutions**.

## INDEPENDENT SET

1. Solve  $|x + 4| = 9$  and find the sum of the roots.
2. How many solutions does  $|3x - 6| = 0$  have?
3. How many integers satisfy  $|x - 3| \leq 5$ ?
4. How many integer solutions does  $|x + 2| + |x - 4| = 6$  have?
5. Solve  $|x - 1| = |x - 7|$ .
6. What is the least value of  $|x - 2| + |x - 6| + |x - 9|$ ?
7. Find the least positive integer solution of  $|5 - 2x| > 3$ .
8. How many solutions does  $||x - 1| - 3| = 2$  have?
9. ★ How many solutions does  $|x - 1| + |x - 3| = x$  have?

## Solutions to the independent sets

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### LESSON 1.1

1.  $0.06x = 12$ , so  $x = \mathbf{200}$ .
  2.  $0.5 \cdot t = 1$ , so  $t = 2$ : a rise of **100%**. Percents down and up are not symmetric, because the bases differ.
  3. Pure acid:  $0.2 + 0.6 = 0.8$  L out of 5 L: **16%**. This is the weighted average of 10 and 20 with weights 2 : 3.
  4.  $b = 12$ : then  $a = 8$ ,  $c = 15$ . Answer: **8 : 15**.
  5. Before: 105; after:  $16 \cdot 8 = 128$ . The number added is **23**.
  6.  $2 \cdot 30 \cdot 60 / (30 + 60) = \mathbf{40 \text{ mph}}$ .
  7. 88 ft/s. Since  $60 \text{ mph} = 88 \text{ ft/s}$ , the speed is **60 mph**.
  8.  $1.1 \cdot 1.1 = 1.21$ : by **21%**. The extra 1% is “percent on percent”: ten percent of ten.
- ★ Each replacement multiplies the amount of acid by  $3/4$ :  $40\% \cdot (3/4)^2 = \mathbf{22.5\%}$ .

### LESSON 1.2

1.  $5x - 10 = 3x + 8$ , so  $2x = 18$ ,  $x = \mathbf{9}$ .
  2.  $2x + 6 = 40$ ,  $x = 17$ : the numbers are 17 and 23, product **391**.
  3.  $1/4 + 1/12 = 1/3$  of the fence per hour, so **3 hours**.
  4. 1.2 L of substance out of 6 L: **20%**.
  5.  $7 + 24 \cdot 4 = \mathbf{103}$ .
  6.  $(1 + 59)/2 \cdot 30 = \mathbf{900}$ . A nice fact: the sum of the first  $n$  odd positive integers is always  $n^2$ .
  7.  $r^2 = 162/18 = 9$ . Then  $a_1 = 18/r^2 = \mathbf{2}$  (the sign of  $r$  does not matter: we divide by  $r^2$ ).
  8. Maya has worked  $t$  minutes, Ben  $t - 5$ : the equation  $2t = 3(t - 5)$  gives  $t = \mathbf{15 \text{ minutes}}$ . Check: both have solved 30 problems.
- ★  $S_{10} = 320$ ,  $S_9 = 261$ : the tenth term is **59**. The trick works for any sum formula: a term is the difference of consecutive sums.

### LESSON 1.3

1.  $x(x - 5) = 0$ : roots **0 and 5**. Dividing by  $x$  loses the root 0.
  2.  $(x - 5)(x + 3) = 0$ : roots 5 and  $-3$ . The largest is **5**.
  3.  $(r + s)/rs = 6/4 = \mathbf{3/2}$ .
  4.  $64 - 3 \cdot 1 \cdot 4 = \mathbf{52}$ .
  5. Vertex at  $x = 4$ :  $-16 + 32 - 7 = \mathbf{9}$ .
  6. The sum of the roots is 6, so the other root is **4** (and incidentally  $c = 8$ ).
  7.  $(x - 2)(x - 3) < 0$  for  $2 < x < 3$ . There are **no integers there: 0**.
  8.  $9 - 2 = 7$ . The equation itself:  $x^2 - 6x + 7 = 0$ .
- ★  $(r + 1)(s + 1) = rs + r + s + 1 = -2 + 3 + 1 = \mathbf{2}$ . No need to find the roots: Vieta's formulas give everything.

### LESSON 1.4

1.  $2 \cdot 100 = \mathbf{200}$ .
2.  $7 - 3 = \mathbf{4}$ .

3.  $216 - 18 = \mathbf{198}$ .

4.  $10^6 - 1 = \mathbf{999,999}$ .

5.  $\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{99}{100}$ : everything cancels down to  $\mathbf{1/100}$ .

6.  $343 - 210 = \mathbf{133}$ .

7. Each term equals  $\sqrt{(k+1)} - \sqrt{k}$ ; the telescope gives  $\sqrt{25} - 1 = \mathbf{4}$ .

8.  $(2^5 - 1)(2^5 + 1)/(2^5 - 1) = 2^5 + 1 = \mathbf{33}$ .

★  $(2-1)(2+1)(2^2+1)(2^4+1)(2^8+1) = 2^{16} - 1$ , so the expression equals  $2^{16} = \mathbf{65,536}$ .

## LESSON 1.5

1. The solutions are 5 and  $-13$ ; they are symmetric about  $-4$ , so the sum is  $\mathbf{-8}$ .

2. Only  $3x - 6 = 0$ : **one solution**,  $x = 2$ .

3. From  $-2$  to  $8$  inclusive: **11 integers**.

4. The distance between  $-2$  and  $4$  is  $6$ , so the equation holds on the whole segment  $[-2, 4]$ . Integer solutions: **7**.

5. The midpoint:  $x = \mathbf{4}$ .

6. At  $x = 6$ :  $4 + 0 + 3 = \mathbf{7}$ .

7. Positive integers with  $x > 4$  start at **5** (1 through 4 fail: substitute and check).

8.  $|x - 1| = 5$  gives  $6$  and  $-4$ ;  $|x - 1| = 1$  gives  $2$  and  $0$ . In total, **4 solutions**.

★ For  $x < 1$ :  $4 - 2x = x$ ,  $x = 4/3$  — not in the interval. For  $1 \leq x \leq 3$ :  $2 = x$  — works. For  $x > 3$ :  $2x - 4 = x$ ,  $x = 4$  — works. In total, **2 solutions** ( $2$  and  $4$ ).